

Matrix

Week 1

Lecture 1

n. n. n. Scalar

Scalars : Real num. / จำนวนจริง (+, -, 0)

$$c \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} = \begin{bmatrix} ca_1 & ca_2 \\ ca_3 & ca_4 \end{bmatrix}$$

กฎการคูณ

n. n. n. Matrix

กฎการคูณ

$$\begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \times \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} = \begin{bmatrix} a_1b_1 + a_2b_3 & a_1b_2 + a_2b_4 \\ a_3b_1 + a_4b_3 & a_3b_2 + a_4b_4 \end{bmatrix}$$

ex. 1×2 2×4
เท่ากัน!
= คูณได้!

จำนวนจริง a_{ij} $m \times n$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

m : จำนวน rows
 n : จำนวน column

ex. $\begin{bmatrix} x & y & z \end{bmatrix} 1 \times 3$ $\begin{bmatrix} x \\ y \\ z \end{bmatrix} 3 \times 1$

System

$$\begin{aligned} x - 4y + 3z &= 5 \\ -x + 3y - z &= -3 \\ 2x - 4z &= 6 \end{aligned}$$

constant : ค่าคงที่

$$\begin{bmatrix} 1 & -4 & 3 & 5 \\ -1 & 3 & -1 & -3 \\ 2 & 0 & -4 & 6 \end{bmatrix}$$

Augmented Matrix

$$\begin{bmatrix} 1 & -4 & 3 \\ -1 & 3 & -1 \\ 2 & 0 & -4 \end{bmatrix}$$

constant

Coefficient matrix

n. n. n. Matrix

* ต้องมี size เท่ากัน ($m \times n$ คูณ $n \times m$)

$$\begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} + \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} = \begin{bmatrix} a_1+b_1 & a_2+b_2 \\ a_3+b_3 & a_4+b_4 \end{bmatrix}$$

การบวก

กฎการลบ

$$\begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} - \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} = \begin{bmatrix} a_1-b_1 & a_2-b_2 \\ a_3-b_3 & a_4-b_4 \end{bmatrix}$$

เลขโดด 101 B 0.1 -1 11.2 21 + 5.4

★

$$\begin{aligned} A + B &= B + A \\ A + (B + C) &= (A + B) + C \\ (cA)B &= c(AB) \end{aligned}$$

scalars

$$1A = A$$

$$c(A+B) = cA + cB \quad n: 114$$

$$(c+d)A = cA + dA \quad n: 114 \text{ ได้}$$

★

★

การคูณ

$$\begin{aligned} A(BC) &= (AB)C \\ A(B+C) &= AB + AC \\ (A+B)C &= AC + BC \\ c(AB) &= (cA)B = A(cB) \end{aligned}$$

constant คูณ

$$\begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

เลขโดด 101 B 0.1 -1 11.2 21 + 5.4

$n \times n$: I_n

$$\begin{aligned} AI_n &= A \\ I_n A &= A \end{aligned}$$

Transpose : $\text{đầu } 1100 \rightarrow \text{uốn}$
 $\text{đầu } 0000 \rightarrow 1100$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

Symmetric matrix EX. $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$

$\Rightarrow$ $\begin{bmatrix} x & - & - \\ - & x & - \\ - & - & x \end{bmatrix} \rightarrow \text{màu đỏ: đều}$
 màu xanh: đảo

$\Rightarrow A = A^T$

$(A^T)^T = A$

$(A+B)^T = A^T + B^T$

$(cA)^T = c(A^T)$

$(AB)^T = B^T A^T$

van : ngược lại

lưu : đầu uôi đầu uôi $\rightarrow$ ngược

Lecture 2

Inverse : Matrix đầu Invertible (đầu Inverse)
 $= \text{nonsingular}$ (không kỳ)

$\Rightarrow AB = BA = I_n$

$\therefore B = \text{inverse of } A = A^{-1}$

$I_n = AA^{-1}$

không inverse : noninvertible (singular)

2x2

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

nxn

$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

$(A^{-1})^{-1} = A$

$(A^k)^{-1} = (A^{-1})^k$

$(cA)^{-1} = \frac{1}{c} A^{-1}, c \neq 0$

$(A^T)^{-1} = (A^{-1})^T$

$(AB)^{-1} = B^{-1} A^{-1}$ $\rightarrow$ ngược ngược

n square matrix $\rightarrow$ $\det$

det

$\det(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \in \begin{matrix} a_{11} \cdot a_{22} \\ -a_{21} \cdot a_{12} \end{matrix} = \begin{vmatrix} a_1 & a_2 \\ a_3 & a_4 \end{vmatrix}$

Cofactors $= (-1)^{ij} \times M_{ij}$ Minor

$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

EX. $M_{21} = \text{đầu } 2 \text{ đầu } 1 = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$

$C_{21} = (-1)^{2+1} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$ Cofactors

đầu Minor: M_{ij}

$\begin{bmatrix} +M_{11} & -M_{12} & +M_{13} \\ -M_{21} & +M_{22} & -M_{23} \\ +M_{31} & -M_{32} & +M_{33} \end{bmatrix}$ $\text{đầu } +, -$
 $\text{đầu } = \begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$

$\begin{bmatrix} + & + \\ - & + \\ + & - \end{bmatrix}_{3 \times 3}$ $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}_{4 \times 4}$

$\det(A) = \sum_{j=1}^n a_{ij} C_{ij}$ $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ $\text{đầu } 1 \text{ đầu } 1$
 $\det A = a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13}$

trick $\text{đầu } 1 \text{ đầu } 1$ EX. $0 \ 0 \ a_1$

th $|A|$ là $\begin{vmatrix} 0 & 0 & 0 \\ 2 & 4 & -5 \\ 3 & -5 & 2 \end{vmatrix}, \begin{vmatrix} 1 & -2 & 4 \\ 0 & 1 & 2 \\ 1 & -2 & 4 \end{vmatrix}, \begin{vmatrix} 1 & 2 & -3 \\ 2 & -1 & -6 \\ -2 & 0 & 6 \end{vmatrix} = 0$

(đi 0) (không đổi) (không đổi)

$\det(AB) = \det(A) \det(B)$
 $\det(cA) = c^n \det(A)$
 $\det(A) \neq 0 : A$ là invertible (nonsingular)
 $\det(A^{-1}) = \frac{1}{\det(A)}$
 $\det(A) = \det(A^T)$

* Adjoint = $\text{Adj } A = (\text{Cofactor})^T$ *

* $A^{-1} = \frac{1}{\det A} \text{adj } A$ *

determinant là m.n., di tích v.v

>> m.n. Δ qđ $(x_1, y_1), (x_2, y_2), (x_3, y_3)$

$(x_1, y_1), (x_2, y_2), (x_3, y_3) \rightarrow A = \pm \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

đi qđ là 20

>> m.n. m.v. qđ (x_1, y_1) and (x_2, y_2)

$= \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{bmatrix}$

>> m.n. Tetrahedron (đi: qđ v.v)

$V = \pm \frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix}$

Cryptography

$— = 0$

$A = 1$ mã message $\rightarrow$ mã (uncoded row matrices)

$B = 2$ EX. MEET ME MONDAY mã là mã 3 chữ

$C = 3$ $\rightarrow$ MEET- ME- MONDAY $\rightarrow$ " " = —

$\vdots$

uncoded row matrices 1x3

$Z = AB = \begin{bmatrix} 13 & 5 & 5 \\ 20 & 0 & 13 \\ 9 & 0 & 13 \\ 19 & 14 & 4 \\ 1 & 25 & 0 \end{bmatrix}$

mã

Encode là Matrix A và mã là uncoded row matrices $\rightarrow$ mã coded row matrices m.n. m.v.

EX. $A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$ mã là mã

MEET ME MONDAY = $\begin{bmatrix} 13 & 5 & 5 \\ 20 & 0 & 13 \\ 9 & 0 & 13 \\ 19 & 14 & 4 \\ 1 & 25 & 0 \end{bmatrix}$

$\begin{bmatrix} 13 & 5 & 5 \end{bmatrix} \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} = \begin{bmatrix} 13 & -26 & 21 \end{bmatrix}$

m.n. m.v.

$\begin{bmatrix} 1 & 25 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} = \begin{bmatrix} -24 & 23 & 77 \end{bmatrix}$

mã m.v. (m.n. m.v.) $\begin{bmatrix} 13 & -26 & 21 \end{bmatrix} \dots \begin{bmatrix} -24 & 23 & 77 \end{bmatrix}$

nonsingular

Decode mã là mã là mã

$\text{m.n. } YA^{-1} = \frac{\text{uncoded row matrices}}{\text{coded row matrices}}$

EX. $A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$

decode $\begin{bmatrix} 13 & -26 & 21 \\ 33 & -53 & -12 \\ 19 & -23 & -42 \\ 5 & -20 & 56 \\ -24 & 23 & 77 \end{bmatrix}$

$U_1 A^{-1} = \frac{1}{\det A} \text{Adj } A$ $\det A = \begin{vmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{vmatrix} = \begin{vmatrix} -1 & -1 & 0 \\ -10 & -6 & -1 \\ -9 & -5 & -1 \end{vmatrix}$

mã là mã là mã

$\begin{bmatrix} 13 & -26 & 21 \\ \vdots \\ -24 & 23 & 77 \end{bmatrix} \begin{bmatrix} -1 & -1 & 0 \\ -10 & -6 & -1 \\ -9 & -5 & -1 \end{bmatrix} = \begin{bmatrix} \dots & \dots & \dots \\ \vdots \\ \dots & \dots & \dots \end{bmatrix}$

mã là mã là mã

Lecture 3 Linear Equations

leading variable
 coefficients
 leading coefficient : non-zero

ex. 1.1.1 Linear Equation

non xy (non-linear)

e^x (expo)

$\sin x$ (non-linear) $\rightarrow (\sin \frac{\pi}{2}) x$, for

$\frac{1}{x} \times \frac{1}{y} = \frac{y+x}{xy}$ - non-linear!

System of Linear Equations

= 2 Linear Equations > 1 var

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

- >> มีคำตอบเดียว (consistent sys.)
- >> มีคำตอบ ∞ " " " " " " " "
- >> ไม่มีคำตอบ (inconsistent sys.) : ระบบไม่สอดคล้อง

$$\begin{aligned} x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ z &= 2 \end{aligned}$$

row-echelon form
 if back-substitution.
 non-zero row

transformation matrix for $AX = b$, $X = A^{-1}b$

$$\begin{aligned} \text{If } AC = BD &\rightarrow A = B \\ \text{If } CA = EB &\rightarrow A = B \end{aligned}$$

ex. $2x + 3y + z = -1$
 $3x + 3y + z = 1$
 $2x + 4y + z = -2 \rightarrow b = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 3 & 1 \\ 2 & 4 & 1 \end{bmatrix}$$

$\det A = 1$
 $\text{Adj. } A = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ 6 & -2 & 3 \end{bmatrix}$
 $A^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ 6 & -2 & 3 \end{bmatrix}$

$$X = A^{-1}b = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ 6 & -2 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix} \text{ for } x = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$$

Cramer +

ex. $4x_1 - 2x_2 = 10$
 $3x_1 - 5x_2 = 11$

$$A = \begin{bmatrix} 4 & -2 \\ 3 & -5 \end{bmatrix}$$

$\det(A) = 4(-5) - (-2)(3) = -20 + 6 = -14$

$$x_1 = \frac{\begin{vmatrix} 10 & -2 \\ 11 & -5 \end{vmatrix}}{\det(A)} = \frac{10(-5) - (-2)(11)}{-14} = \frac{-50 + 22}{-14} = \frac{-28}{-14} = 2$$

$$x_2 = \frac{\begin{vmatrix} 4 & 10 \\ 3 & 11 \end{vmatrix}}{\det(A)} = \frac{4(11) - 3(10)}{-14} = \frac{44 - 30}{-14} = \frac{14}{-14} = -1$$

Gaussian +

ex. $x - 2y + 3z = 9$
 $-x + 3y = -4$
 $2x - 5y + 5z = 17$

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ -1 & 3 & 0 & -4 \\ 2 & -5 & 5 & 17 \end{bmatrix} \xrightarrow{R_2+R_1, R_3-2R_1} \begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & -1 & -1 & -1 \end{bmatrix}$$

$$\xrightarrow{R_3+R_2} \begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & 4 \end{bmatrix} \xrightarrow{R_3/2} \begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Row-echelon form
 back-substitution

$$\begin{aligned} z &= 2 \\ y + 3z &= 5 \rightarrow y + 6 = 5 \rightarrow y = -1 \\ x - 2y + 3z &= 9 \rightarrow x - 2(-1) + 6 = 9 \rightarrow x + 2 + 6 = 9 \rightarrow x + 8 = 9 \rightarrow x = 1 \end{aligned}$$

$$\begin{bmatrix} 1 & -5 & 2 & -1 & 3 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

check for

Gauss-Jordan +

for I
 $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \text{identity}$

Polynomial Curve Fitting



ex. points are (1,4), (2,0) and (3,12)

$$p(x) = a_0 + a_1x + a_2x^2$$

$$\begin{aligned} (1,4) &\rightarrow 4 = a_0 + a_1 + a_2 \\ (2,0) &\rightarrow 0 = a_0 + 2a_1 + 4a_2 \\ (3,12) &\rightarrow 12 = a_0 + 3a_1 + 9a_2 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 & 1 & 4 \\ 1 & 2 & 4 & 0 \\ 1 & 3 & 9 & 12 \end{bmatrix} \xrightarrow{R_2-R_1, R_3-R_1} \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & -4 \\ 0 & 2 & 8 & 8 \end{bmatrix}$$

$$\xrightarrow{R_3-2R_2} \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & -4 \\ 0 & 0 & 2 & 16 \end{bmatrix} \xrightarrow{R_3/2} \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & -4 \\ 0 & 0 & 1 & 8 \end{bmatrix}$$

$$\xrightarrow{R_2-3R_3} \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & -28 \\ 0 & 0 & 1 & 8 \end{bmatrix} \xrightarrow{R_1-R_2} \begin{bmatrix} 1 & 0 & 1 & 32 \\ 0 & 1 & 0 & -28 \\ 0 & 0 & 1 & 8 \end{bmatrix}$$

$$\xrightarrow{R_1-R_3} \begin{bmatrix} 1 & 0 & 0 & 24 \\ 0 & 1 & 0 & -28 \\ 0 & 0 & 1 & 8 \end{bmatrix}$$

The LU-factorization $AX = b$

X

$A = LU$

$$L = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad U = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

Lower triangular Matrix Upper triangular Matrix

$$A = LU \text{ nnu } = LUx = b$$

$$\begin{aligned} L(Ux) &= b \\ \text{let } Ux &= y \end{aligned} \quad \begin{cases} Ly = b \\ Ux = y \end{cases}$$

m LU-factorization was $A = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{bmatrix}$

$$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & -4 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix}$$

Vectors in $\mathbb{R}^n$

$$\mathbb{R}^2 = \langle x_1, x_2 \rangle$$

$$\mathbb{R}^3 = \langle x_1, x_2, x_3 \rangle$$

$$\mathbb{R}^n = \langle x_1, x_2, \dots, x_n \rangle$$

$$0\vec{v} = \vec{0}$$

$$c\vec{0} = \vec{0}$$

$$c\vec{v} = \vec{0} \rightarrow c=0 \text{ or } \vec{v} = \vec{0}$$

$$-(\vec{v}) = \vec{v}$$

Length $\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$

$$\|c\vec{v}\| = |c| \|\vec{v}\|$$

Dot $\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + \dots + u_nv_n$

$$\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$$

$$\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$$

$$c(\vec{u} \cdot \vec{v}) = (c\vec{u}) \cdot \vec{v} = \vec{u} \cdot (c\vec{v})$$

$$\vec{v} \cdot \vec{v} = \|\vec{v}\|^2$$

$$\vec{v} \cdot \vec{v} \geq 0, \vec{v} \cdot \vec{v} = 0 \iff \vec{v} = \vec{0}$$

ex. $\vec{u} \cdot \vec{u} = \vec{u}^2 = \dots$

$$\vec{u} \cdot \vec{v} = \dots$$

$$\vec{v} \cdot \vec{v} = \dots$$

$(u+2v) \cdot (3u+v) = ?$

$$3u^2 + 7u \cdot v + 2v^2$$

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|} \rightarrow \vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$$

$$\cos \theta = -1 \rightarrow \theta = \pi$$

$$\cos \theta < 0 \rightarrow \theta \in (\pi/2, \pi)$$

$$\cos \theta = 0 \rightarrow \theta = \pi/2$$

$$\cos \theta > 0 \rightarrow \theta \in (0, \pi/2)$$

$$\cos \theta = 1 \rightarrow \theta = 0$$

Orthogonal Vector.

$$\vec{u} \cdot \vec{v} = 0$$

$$|\vec{u} \times \vec{v}| = |\vec{u}| |\vec{v}| \sin \theta = \text{Area}$$

base height



Parallellogram

$$\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$$

$$\vec{u} \times (\vec{v} + \vec{w}) = (\vec{u} \times \vec{v}) + (\vec{u} \times \vec{w})$$

$$c(\vec{u} \times \vec{v}) = c\vec{u} \times \vec{v} = \vec{u} \times c\vec{v}$$

$$\vec{u} \times \vec{0} = \vec{0}$$

$$\vec{u} \times \vec{u} = \vec{0}$$

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = (\vec{u} \cdot \vec{v}) \cdot \vec{w}$$

Unit Vector

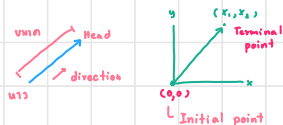
$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$$

Lecture 4 Vector spaces

Scalar : $\vec{v}$ scalar

vector : $\vec{v}$ scalar + vector

row or column



Vector $\vec{v} = \langle v_1, v_2 \rangle$, $\vec{u} = \langle u_1, u_2 \rangle$

$$\vec{v} = \langle v_1, v_2 \rangle, \vec{u} = \langle u_1, u_2 \rangle$$

Unit Vector

$$\vec{u} + \vec{v} = \langle u_1 + v_1, u_2 + v_2 \rangle$$

Unit Vector Scalar : $c\vec{v} = \langle cv_1, cv_2 \rangle$

$$c > 0 \text{ then } \vec{v} \text{ is scaled up}$$

$$c < 0 \text{ then } \vec{v} \text{ is scaled down}$$

Lecture 5

Vector Spaces / သံသရာ

1. $\vec{u} + \vec{v}$ is in $\underline{V}$

2. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

3. $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$

4. V has zero vector 0

၇၈၈ $\vec{u}$ in V , $\vec{u} + \vec{0} = \vec{u}$ လာရက်

5. $\vec{u}$ in V $\vec{u} + (-\vec{u}) = 0$ ဝါရက်

$-\vec{u}$ is $\vec{u} + (-\vec{u}) = 0$ ဝါရက်

6. $c\vec{u}$ is in $\underline{V}$

7. $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$

8. $(c+d)\vec{u} = c\vec{u} + d\vec{u}$

9. $c(d\vec{u}) = (cd)\vec{u}$

10. $(c\vec{u}) = \vec{u}$ လာရက်

1. $0\vec{v} = \vec{0}$

2. $c\vec{0} = \vec{0}$

3. $c\vec{v} = \vec{0} \rightarrow c = 0$ or $\vec{v} = \vec{0}$

4. $(-1)\vec{v} = -\vec{v}$

ex. $\vec{u} = x^2$ Degree 2

$\vec{u} = x^2$, $\vec{v} = x$

$\vec{u} + \vec{v} = x^2 + x$

$\vec{u} + \vec{v} = x^2 + x$ degree 2 လာရက်

$d = R$

ex. $\vec{u} = 1$

$\vec{u} = 1$

$c = \frac{1}{2}$

$c\vec{u} = \frac{1}{2}$ လာရက်

ex. $V = R^2$

$c(\vec{x}_1, \vec{x}_2) = (c\vec{x}_1, c\vec{x}_2)$

$c(\vec{u}) = \vec{u}$

$\vec{u} = x_1, x_2, 1 \rightarrow (1, 1)$

$c(1, 1) = (c, c) \neq (1, 1)$

Sub spaces : proof no 1, 6

ex. $V = R^3$

$\vec{u} = (x_1, 0, x_3)$

$\vec{u} = (y_1, 0, y_3)$ ဝါရက်

$\vec{u} + \vec{v} = (x_1 + y_1, 0, x_3 + y_3)$

$\vec{u} + \vec{v} = (x_1 + y_1, 0, x_3 + y_3)$ ဝါရက်

$\vec{u} = c\vec{u}$ ဝါရက်

$c(x_1, 0, x_3) = (cx_1, 0, cx_3)$ ဝါရက်

= ဝါရက် Subspace

→ လာရက် ex. Singular matrices ufo လာရက် Inverse.

$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

$A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\det \neq 0 \therefore$ inverse.

ex. Quadrant 1 $\therefore (+, +)$

Lo $\vec{u} = (1, 1)$ ဝါရက်

$c = -1 \rightarrow c\vec{u} = -(1, 1) = (-1, -1)$

လာရက်

Linear Combination

thông số

$$v = c_1 \vec{u}_1 + c_2 \vec{u}_2 + \dots + c_n \vec{u}_n$$

ex. $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -5 \end{pmatrix} \right\}$

V_1 là linear combination của V_2, V_3

thì: $V_1 = 3V_2 + V_3$

ex. show $w = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ là linear combination

cho $S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$

$$w = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = (c_1 - c_3, 2c_1 + c_2, 3c_1 + 2c_2 + c_3)$$

$$\therefore c_1 - c_3 = 1$$

$$2c_1 + c_2 = 1$$

$$3c_1 + 2c_2 + c_3 = 1$$

$$\rightarrow \begin{bmatrix} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & 1 \\ 0 & 1 & 2 & -1 \end{bmatrix} \quad R_3 - R_1 - R_2$$

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 2 & -1 \end{bmatrix} \quad R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_3 - R_2$$

$$\rightarrow c_1 - c_3 = 1 \rightarrow c_1 = c_3 + 1$$

$$c_2 + 2c_3 = -1 \rightarrow c_2 = -1 - 2c_3$$

$$\text{let } c_3 = t, \quad c_1 = t + 1$$

$$c_2 = -1 - 2t$$

$$\text{let } t = 1 \text{ thì } c_1 = 2, c_2 = -3, c_3 = 1$$

$$\therefore w = 2V_1 - 3V_2 + V_3$$

Spanning Sets

nếu Vector $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ là $(c_1, c_2, \dots, c_n)$

$$\text{thì } v = c_1 u_1 + c_2 u_2 + c_3 u_3 + \dots + c_n u_n$$

thì ... thì

ex. $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ spans $\mathbb{R}^3$ với 3

thì: $\vec{u} = (u_1, u_2, u_3)$

$$\rightarrow \vec{u} = u_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + u_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + u_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = (u_1, u_2, u_3)$$

ex. show $S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} \right\}$ spans $\mathbb{R}^3$

$$(u_1, u_2, u_3) = (c_1 - 2c_3, 2c_1 + c_2, 3c_1 + 2c_2 + c_3)$$

$$\text{let det } \begin{vmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix} \begin{vmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{vmatrix}$$

$$1 - 6 + 6 = -1 \quad \checkmark$$

$\det(A) \neq 0$: nếu là 0 thì không thể

၈. ပုံစံသတ်မှတ်ပါ

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n = \vec{0} \quad \left\{ \begin{array}{l} \text{ပုံစံသတ်မှတ်ပါ} \\ \text{ပုံစံသတ်မှတ်ပါ} \end{array} \right.$$

ex. $S = \{(1, 2), (2, 4)\} \subset \mathbb{R}^2$

$$-2(1, 2) + (2, 4) = (0, 0)$$

$\hookrightarrow c_1 \neq 0, c_2 \neq 0 \therefore$ ပုံစံသတ်မှတ်ပါ

ex. $S = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\} \subset \mathbb{R}^3$ ပုံစံသတ်မှတ်ပါ

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + c_3 \vec{v}_3 = \vec{0}$$

$$(c_1 - 2c_3, c_2 + c_3, 3c_1 + 2c_2 + c_3) = (0, 0, 0)$$

သို့သော် c_1, c_2, c_3

$$\begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \text{row 1} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$\therefore c_1 = c_2 = c_3 = 0$

$\therefore$ ပုံစံသတ်မှတ်ပါ

Lecture 6

preimage $T: V \rightarrow W$ $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$
image $T(\vec{v}) = A\vec{v}$

ex. $TC(v_2) = C v_1 - v_2 + v_1 + 2v_2$

$\gg$ image $\vec{v}: (-1, 2)$

for $\vec{v}$ image $T(-1, 2) = (-1, -2, -1+4)$

$\therefore$ image $\rightarrow (3, 3) \neq$

$\gg$ preimage $\vec{w}: (-1, 11)$

$$(-1, 11) = (v_1 - v_2, v_1 + 2v_2)$$

$$\left. \begin{array}{l} v_1 - v_2 = -1 \\ v_1 + 2v_2 = 11 \end{array} \right\} \rightarrow \begin{array}{l} 3v_2 = 12 \\ v_2 = 4 \\ \therefore v_1 = 3 \end{array}$$

$$T(c\vec{u} + \vec{v}) = T(c\vec{u}) + T(\vec{v})$$

$$T(c\vec{u}) = c(T\vec{u})$$

Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation such that

$$T(1, 0, 0) = (2, -1, 4)$$

$$T(0, 1, 0) = (1, 5, -2)$$

$$T(0, 0, 1) = (0, 3, 1)$$

$$\text{Find } T(2, 3, -2)$$

$$T(2, 3, -2) = 2T(1, 0, 0) + 3T(0, 1, 0) - 2T(0, 0, 1)$$

$$= 2(2, -1, 4) + 3(1, 5, -2) - 2(0, 3, 1)$$

$$= (4, -2, 8) + (3, 15, -6) - (0, 6, 2)$$

$$= (7, 7, 0) \neq$$

The function $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is defined as follows.

$$T(\vec{v}) = A\vec{v} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

(a) Find $T(\vec{v})$, where $\vec{v} = (2, -1)$.

(b) Show that T is a linear transformation from $\mathbb{R}^2$ into $\mathbb{R}^3$.

$$T(\vec{v}) = \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 3v_1 \\ 2v_1 + v_2 \\ -v_1 - 2v_2 \end{bmatrix}$$

$$T(c\vec{u} + \vec{v}) = (3, 5, 0)$$

$$T(c\vec{u}) + T(\vec{v}) =$$

$$T(c\vec{u}) + T(\vec{v}) =$$

$$T(c\vec{u}) = ?$$

The linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is defined by $T(\vec{v}) = A\vec{v}$. Find the dimensions of $\mathbb{R}^n$ and $\mathbb{R}^m$ for the linear transformation represented by each matrix.

(a) $A = \begin{bmatrix} 0 & 1 & -1 \\ 2 & 3 & 0 \\ 4 & 2 & 1 \end{bmatrix}$ $\rightarrow \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \rightarrow \begin{bmatrix} v_1 - v_2 \\ 2v_1 + 3v_2 \\ 4v_1 + 2v_2 + v_3 \end{bmatrix}$

(b) $A = \begin{bmatrix} 2 & -3 \\ -5 & 0 \\ 0 & -2 \end{bmatrix}$ $\rightarrow \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \rightarrow \begin{bmatrix} 2v_1 - 3v_2 \\ -5v_1 \\ -2v_2 \end{bmatrix}$

(c) $A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 3 & 1 & 0 & 0 \end{bmatrix}$ $\rightarrow \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} \rightarrow \begin{bmatrix} v_1 - v_3 + 2v_4 \\ 3v_1 + v_2 \end{bmatrix}$

$$T(\vec{u}) = A\vec{u} \rightarrow \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} \rightarrow \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

The linear transformation $T: R^3 \rightarrow R^3$ represented by

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

is called a **projection** in R^3 . If $v = (x, y, z)$ is a vector into R^3 , then $T(v) = (x, y, 0)$. In other words, T maps every vector in R^3 to its orthogonal projection in the xy -plane, as shown in Figure 6.3.

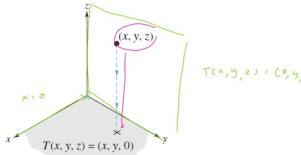


Figure 6.3

Let $T: M_{m,n} \rightarrow M_{m,n}$ be the function that maps an $m \times n$ matrix A to its transpose. That is,

$$T(A) = A^T$$

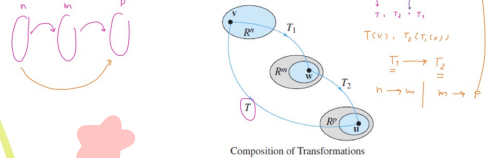
Show that T is a linear transformation.

1. $T(A+B) = (A+B)^T = A^T + B^T = T(A) + T(B)$

2. $T(cA) = (cA)^T = cA^T = cT(A)$

Composition of Linear Transformations

Let $T_1: R^n \rightarrow R^m$ and $T_2: R^m \rightarrow R^p$ be linear transformations with standard matrices A_1 and A_2 . The **composition** $T: R^n \rightarrow R^p$, defined by $T(v) = T_2(T_1(v))$, is a linear transformation. Moreover, the standard matrix A for T is given by the matrix product $A = A_2 A_1$.



Let T_1 and T_2 be linear transformations from R^3 into R^3 such that

$$T_1(x, y, z) = (2x + y, 0, x + z) \quad \text{and} \quad T_2(x, y, z) = (x - y, z, y)$$

Find the standard matrices for the compositions $T = T_2 \circ T_1$ and $T' = T_1 \circ T_2$.

Find standard matrices for T_1 and T_2 :

$$T_1(v) = \begin{bmatrix} 2x+y \\ 0 \\ x+z \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow A_1 = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$T_2(v) = \begin{bmatrix} x-y \\ z \\ y \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow A_2 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Then $A = A_2 A_1$ and $A' = A_1 A_2$.

Definition of Inverse Linear Transformation

If $T_1: R^n \rightarrow R^n$ and $T_2: R^n \rightarrow R^n$ are linear transformations such that for every v in R^n

$$T_2(T_1(v)) = v \quad \text{and} \quad T_1(T_2(v)) = v$$

then T_2 is called the **inverse** of T_1 , and T_1 is said to be **invertible**.

Not every linear transformation has an inverse. If the transformation T_1 is invertible, however, then the inverse is unique and is denoted by T_1^{-1} . Just as the inverse of a function of a real variable can be thought of as undoing what the function did, the inverse of a linear transformation T can be thought of as undoing the mapping done by T . For instance, if T is a linear transformation from R^3 onto R^3 such that

$$T(1, 4, -5) = (2, 3, 1)$$

and if T^{-1} exists, then T^{-1} maps $(2, 3, 1)$ back to its preimage under T . That is,

$$T^{-1}(2, 3, 1) = (1, 4, -5).$$

Existence of an Inverse Transformation

Let $T: R^n \rightarrow R^n$ be a linear transformation with standard matrix A . Then the following conditions are equivalent.

- 1. T is invertible.
 - 2. T is an isomorphism.
 - 3. A is invertible.
- And, if T is invertible with standard matrix A , then the standard matrix for T^{-1} is A^{-1} .

The linear transformation $T: R^2 \rightarrow R^2$ is given by projecting each point in R^2 onto the x -axis, as shown in Figure 6.8. Find the standard matrix for T .

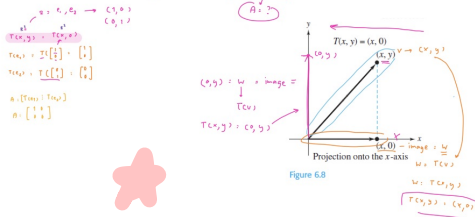


Figure 6.8

Finding the Inverse of a Linear Transformation

The linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by

$$T(x_1, x_2, x_3) = (2x_1 + 3x_2 + x_3, 3x_1 + 3x_2 + x_3, 2x_1 + 4x_2 + x_3).$$

Show that T is invertible, and find its inverse.

$E \rightarrow 3$

$E_1 = (1, 0, 0)$
 $E_2 = (0, 1, 0)$
 $E_3 = (0, 0, 1)$

$A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 3 & 1 \\ 2 & 4 & 1 \end{bmatrix}$

$T^{-1} = A^{-1}V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$T^{-1}(x_1, x_2, x_3) = (x_1, x_2, x_3)$

Elementary Matrices for Linear Transformations in the Plane

Reflection in y -Axis
 $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

Reflection in x -Axis
 $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Reflection in Line $y = x$
 $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Horizontal Expansion ($k > 1$) or Contraction ($0 < k < 1$)
 $A = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$

Vertical Expansion ($k > 1$) or Contraction ($0 < k < 1$)
 $A = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$

Horizontal Shear
 $A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$

Vertical Shear
 $A = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$

Lecture 7

Eigenvalues and Eigenvectors of a Matrix

Let A be an $n \times n$ matrix.

1. An eigenvalue of A is a scalar λ such that

$$\det(A - \lambda I) = 0.$$

2. The eigenvectors of A corresponding to λ are the nonzero solutions of

$$(A - \lambda I)x = 0.$$

The equation $\det(A - \lambda I) = 0$ is called the **characteristic equation** of A . Moreover, when expanded to polynomial form, the polynomial

$$|A - \lambda I| = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_1\lambda + c_0$$

is called the **characteristic polynomial** of A . This definition tells you that the eigenvalues of an $n \times n$ matrix A correspond to the roots of the characteristic polynomial of A . Because the characteristic polynomial of A is of degree n , A can have at most n distinct eigenvalues.

REMARK: The Fundamental Theorem of Algebra states that an n th-degree polynomial has precisely n roots. These n roots, however, include both repeated and complex roots. In this chapter you will be concerned only with the real roots of characteristic polynomials—that is, real eigenvalues.

Finding Eigenvalues and Eigenvectors

Find the eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix}$$

SOLUTION The characteristic polynomial of A is

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 2 - \lambda & -12 \\ 1 & -5 - \lambda \end{vmatrix} = (2 - \lambda)(-5 - \lambda) - (-12) \\ &= (\lambda - 2)(\lambda + 5) - (-12) \\ &= \lambda^2 + 3\lambda - 10 + 12 \\ &= \lambda^2 + 3\lambda + 2 \\ &= (\lambda + 1)(\lambda + 2) = 0 \end{aligned}$$

So, the characteristic equation is $(\lambda + 1)(\lambda + 2) = 0$, which gives $\lambda_1 = -1$ and $\lambda_2 = -2$ as the eigenvalues of A . To find the corresponding eigenvectors, use Gauss-Jordan elimination to solve the homogeneous linear system represented by $(A - \lambda I)x = 0$ twice: first for $\lambda_1 = -1$, and then for $\lambda_2 = -2$. For $\lambda_1 = -1$, the coefficient matrix is

$$(-1)I - A = \begin{bmatrix} -1 - 2 & 12 \\ -1 & -1 + 5 \end{bmatrix} = \begin{bmatrix} -3 & 12 \\ -1 & 4 \end{bmatrix}$$

which row reduces to

$$\begin{bmatrix} 1 & -4 \\ 0 & 0 \end{bmatrix}$$

showing that $x_1 - 4x_2 = 0$. Letting $x_2 = t$, you can conclude that every eigenvector of A_1 is of the form

$$\begin{bmatrix} 4t \\ t \end{bmatrix} = t \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad t \neq 0.$$

For $\lambda_2 = -2$, you have

$$(-2)I - A = \begin{bmatrix} -2 - 2 & 12 \\ -2 & -5 + 2 \end{bmatrix} = \begin{bmatrix} -4 & 12 \\ -2 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 \\ 0 & 0 \end{bmatrix}$$

Letting $x_2 = t$, you can conclude that every eigenvector of A_2 is of the form

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3t \\ t \end{bmatrix} = t \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad t \neq 0.$$

Try checking $Ax = \lambda x$ for the eigenvalues and eigenvectors in this example.

Find the eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

What is the dimension of the eigenspace of each eigenvalue?

SOLUTION

The characteristic polynomial of A is

$$|\lambda I - A| = \begin{vmatrix} \lambda - 2 & -1 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 2 \end{vmatrix} = (\lambda - 2)^3.$$

So, the characteristic equation is $(\lambda - 2)^3 = 0$.

So, the only eigenvalue is $\lambda = 2$. To find the eigenvectors of $\lambda = 2$, solve the homogeneous linear system represented by $(2I - A)x = 0$.

$$2I - A = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

This implies that $x_2 = 0$. Using the parameters $s = x_1$ and $t = x_3$, you can find that the eigenvectors of $\lambda = 2$ are of the form

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} s \\ 0 \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad s \text{ and } t \text{ not both zero.}$$

Because $\lambda = 2$ has two linearly independent eigenvectors, the dimension of its eigenspace is 2.

Find the eigenvalues of

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -10 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{bmatrix}$$

and find a basis for each of the corresponding eigenspaces.

SOLUTION The characteristic polynomial of A is

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} \lambda - 1 & 0 & 0 & 0 \\ 0 & \lambda - 1 & -5 & 10 \\ -1 & 0 & \lambda - 2 & 0 \\ -1 & 0 & 0 & \lambda - 3 \end{vmatrix} \\ &= (\lambda - 1)^2(\lambda - 2)(\lambda - 3). \end{aligned}$$

So, the characteristic equation is $(\lambda - 1)^2(\lambda - 2)(\lambda - 3) = 0$ and the eigenvalues are

$\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = 3$. (Note that $\lambda_1 = 1$ has a multiplicity of 2.)

You can find a basis for the eigenspace of $\lambda_1 = 1$ as follows.

$$(1)I - A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -5 & 10 \\ -1 & 0 & -1 & 0 \\ -1 & 0 & 0 & -2 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} -1 & 0 & 0 & -2 \\ 0 & 0 & -5 & 10 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Letting $s = x_2$ and $t = x_4$ produces

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ s \\ 0 \\ t \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 0 \\ 0 \\ 1 \end{bmatrix} = s \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

A basis for the eigenspace corresponding to $\lambda_1 = 1$ is

$$B_1 = \{(0, 1, 0, 0), (-2, 0, 2, 1)\}.$$

For $\lambda_2 = 2$ and $\lambda_3 = 3$, follow the same pattern to obtain the eigenspace bases

$$B_2 = \{(0, 5, 1, 0)\}$$

$$B_3 = \{(0, -5, 0, 1)\}.$$

Find the eigenvalues of each matrix.

$$(a) A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ 5 & 3 & -3 \end{bmatrix} \quad (b) A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$$

SOLUTION (a) Without using Theorem 7.3, you can find that

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} \lambda - 2 & 0 & 0 \\ 1 & \lambda - 1 & 0 \\ -5 & -3 & \lambda + 3 \end{vmatrix} \\ &= (\lambda - 2)(\lambda - 1)(\lambda + 3). \end{aligned}$$

So, the eigenvalues are $\lambda_1 = 2$, $\lambda_2 = 1$, and $\lambda_3 = -3$, which are simply the main diagonal entries of A .

(b) In this case, use Theorem 7.3 to conclude that the eigenvalues are the main diagonal entries $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 0$, $\lambda_4 = 4$, and $\lambda_5 = 3$.

Eigenvalues and Eigenvectors of Linear Transformations

This section began with definitions of eigenvalues and eigenvectors in terms of matrices. They can also be defined in terms of linear transformations. A number λ is called an **eigenvalue** of a linear transformation $T: V \rightarrow V$ if there is a nonzero vector $\mathbf{x}$ such that $T(\mathbf{x}) = \lambda\mathbf{x}$. The vector $\mathbf{x}$ is called an **eigenvector** of T corresponding to λ , and the set of all eigenvectors of λ (with the zero vector) is called the **eigenspace** of λ .

Consider the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, whose matrix relative to the standard basis is

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

Standard basis:

$$B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

In Example 5 of Section 6.4, you found that the matrix of T relative to the basis B' is the diagonal matrix

$$A' = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

Nonstandard basis:

$$B' = \{(1, 1, 0), (1, -1, 0), (0, 0, 1)\}$$

The question now is: "For a given transformation T , can you find a basis B' whose corresponding matrix is diagonal?" The next example gives an indication of the answer.

